Round 2: Deep Third-Cumulant Representations under Hard FLOP Budget (n=1024, L=16, B=2^{41} FLOPs)

Only proposals outside the closed list of §3. Claims labelled D=derived, C=conjecture, M=measure-this. Cost unit: 1 unit = one dense 1024^3 matmul = 2^{31} FLOPs; B = 1024 units.

Q1. Concrete fitting procedure for the Aggregate Response State

Mathematical object (after the age-4 gate). Carry k independent components, each a pair ($G_a \in \mathrm{Sym}_n$, $v_a \in \mathbb{R}^n$), a=1…k. The aggregate third-cumulant contribution of all old sources is modelled as

$$\kappa_3_\mathrm{old} \approx \Sigma_{a=1}^{k} \mathrm{sym3}(G_a \otimes v_a) \equiv \Sigma_a (G_a \otimes v_a + G_a \otimes v_a \text{ (perm.)} + v_a \otimes G_a),$$

so the needed slices are exact linear combinations of element-wise products of the components:

$$D21_ic = \Sigma_a [(G_a)_{ii} (v_a)_c + 2 (G_a)_{ic} (v_a)_i], \quad D3_i = 3 \Sigma_a (G_a)_{ii} (v_a)_i.$$

Per-layer transport (exact under the linear map B = W diag(w_1))

$G_a \leftarrow B G_a B^T$, $v_a \leftarrow B v_a$. Each component costs two dense n×n multiplies (≈ 2 units). Total for the deep tier: 2k units per layer. After transport the new w_2 of the arriving layer is folded into the G's by a diagonal scaling (still $O(n^2)$ and free inside the unit count).

Fitting procedure at the age gate (objective = 2 % relative RMS on the four query families U)

At the moment a source reaches age 4 its dense legs A, P are still available. The exact contribution of that source to the four contracted queries that determine D21/D3 is the set of matrices

$$U_1 = (A \circ P)\, \mathrm{diag}(w_2)\, A^T,\ U_2 = (A \circ A)\, \mathrm{diag}(w_2)\, P^T,\ U_3 = (P \circ P)\, M^T,\ U_4 = (M \circ P)\, P^T \ (M \text{ the third residual leg}).$$

Concatenate them (or their action on a fixed set of 64 random probe vectors) into a single n×m matrix U_exact (m ≈ 4·64 = 256). We seek rank-k factors that reproduce U_exact to 2 % relative Frobenius error while obeying the structural constraint that each factor is of the form G⊗v.

Algorithm (alternating least-squares, D from standard CP-ALS adapted to the sym3 structure):
1. Initialise v_a by the top-k left singular vectors of U_exact; set $G_a \leftarrow 0$.
2. For each a, solve the linear least-squares problem for the free entries of the symmetric G_a given all other components fixed (normal equations cost $O(n^2 m)$ once per a).
3. For each a, solve for v_a given the G's (another $O(n^2 m)$ system).
4. Repeat 2–3 until relative residual stagnates (typically 8–12 iterations). After all sources that cross the gate in a given layer have been fitted, sum the G's and v's of identical "modes" (or simply keep them as a larger pool and prune later by energy).

Cost of the fit itself (once per source at the gate): $\approx 12 \cdot k \cdot (n^2 m / 2^{31}) \approx 0.4$ k units per source. With ≤ 12 sources crossing the gate over the whole network the total fitting overhead is < 5k units — negligible compared with the subsequent transport cost.

Expected k from M1–M3 (C)

M3 states that the exact contracted object U is compressible to rank ≈ 230 for 2 % energy. Because each (G,v) pair contributes a rank-at-most-n factor of a very special form, the effective degrees of freedom per component are only $n(n+1)/2 + n \approx n^2/2$. Empirical CP rank of a generic sym3 tensor that matches a rank-r matrix query is typically 1.5–2 r. Hence the conjecture k ≈ 300–450. At k = 350 the per-layer transport cost is 700 units — already larger than the entire budget B. Even an optimistic k = 80 (if the query rank of M3 overestimates the true structural rank) still costs 160 units per layer, i.e. > B after only a few layers. Thus the aggregate-response idea, while mathematically elegant, is almost certainly too expensive unless a dramatic further compression of the components themselves is discovered.

First falsifier (one layer, one network, stored legs as .npz)

Load the dense legs of all sources of age exactly 4 at layer 6 (or 7). Form the four exact U matrices. Run the ALS procedure above for k = 50, 100, 200, 300. Record relative Frobenius residual on U and the resulting relative RMS error on D21. If residual > 2 % already at k = 200 the method cannot meet the error law at affordable cost; if it succeeds only for k ≥ 250 the transport cost exceeds B. Runtime on a single CPU core with BLAS: < 20 min for the whole sweep.

Q2. Formulations that are not a sum over sources

Every proposal below is required to stay outside the closed list (no per-source dense legs after age 4, no mean-field replacement forbidden by T1/T2, no low-rank sketch of Hadamard products forbidden by M1/M3, no pure even statistics).

(A) Quenched two-point statistics of κ_3 plus a small odd residual basis (C)

Because κ_3 is odd under sign flips of W, its expectation vanishes, but its second moment $E[\kappa_3 \otimes \kappa_3]$ is even and, by rotational invariance and the known Gaussian measure on the weights, is a linear combination of products of the current covariance C. Concretely the only isotropic tensors of the correct symmetry are built from C⊗C⊗C with a handful of distinct index contractions. One can therefore carry, for free, the exact second-moment structure of the deep third cumulant. The realized (quenched) odd part that actually contributes to D21 is then represented by a single low-rank factorisation of size n × r_odd with r_odd ≪ n (the "realized odd residual"). Update: the second-moment tensors are functions of C alone and cost nothing extra; the residual factor is transported by the same $B = W \operatorname{diag}(w_1)$ at cost $2\, n^2\, r_{odd} \approx 0.4$ units for r_odd = 64. Slices are read off by contracting the residual against itself with the known second-moment kernel. Loss: any higher-order cumulants of κ_3 itself. Falsifier: at layer 10 compute the exact deep κ_3 contribution to D21; form the residual after subtracting the best rank-64 approximation consistent with the second-moment kernel; measure relative error. If > 3 % the residual rank is insufficient.

(B) Tensor-network contraction that eliminates the birth index first (D/C)

A single source's contribution is $\kappa_3_ijk = \Sigma_m\, A_im\, A_jm\, P_km\, w_2_m$ (plus permutations). The birth index m can be contracted before any later layer is applied, producing three n×n matrices already at birth: $S_AA = A \operatorname{diag}(w_2)\, A^T$, $S_AP = A \operatorname{diag}(w_2)\, P^T$, $S_PP = P \operatorname{diag}(w_2)\, P^T$. Thereafter only these three matrices need to be transported: $S \leftarrow B\, S\, B^T$. Cost per source after birth: 6 units per layer (three independent covariance-style transports). For the whole old tier this is still 60+ units — better than 107 but not yet at the leaders' 0. Aggregation across sources is free (just sum the S's). The method is exact for every individual source; the only question is whether the summed S's remain full-rank or can themselves be low-rank compressed. Falsifier: form the summed S_AA, S_AP, S_PP of all sources older than age 4 at layer 8; compute their singular spectra. If 99 % energy already requires > 400 directions the subsequent transport stays expensive.

(C) Response-function / Jacobian statistics (C)

The final mean E[h^(L)] is a functional of the initial activation and of every intermediate weight. Its variation with respect to a deep-layer pre-activation can be written as a product of Jacobians of the subsequent ReLU maps. The third-order response (the object that multiplies κ_3) is exactly the third derivative of that composition. Because the composition is a chain of element-wise nonlinearities interleaved with linear maps, the third response tensor admits a recursive update that only needs the current first- and second-order response tensors (size n and n×n) and the diagonal Hermite coefficients. Cost: a few dense n×n multiplies per layer (≈ 3–4 units total for the whole deep response). The method never materialises κ_3; it only ever multiplies the arriving D21/D3 slices by the already-computed response. Falsifier: implement the third-order response recursion on one network, inject the exact deep D21 at a single intermediate layer, and verify that the final-mean perturbation matches the finite-difference sensitivity map of M2 to relative 1 %. Any discrepancy kills the recursion.

Characteristic-function / random-feature expansions of the full layer map were considered and rejected: a low-dimensional feature basis cannot capture the flat, noise-like spectrum of the Hadamard products (M1) while remaining cheaper than forming those products. Spin-glass TAP / cavity expansions that keep only diagonal or mean-field odd moments are forbidden by T1 (the conditional expectation vanishes). Feynman-diagram resummations that stay at $O(n^2)$ recover at best the covariance + κ_4 regeneration already present in V29 and cannot see the cross-neuron third cumulant.

Q3. Most likely explanation of the leaders' cost/accuracy

Hypothesis (C, ranked first): the leaders keep the young-source tier exactly as V29 does (≈ 140–155 units) and replace the entire old-source tier by a single dense n×n "effective D21 accumulator" that is updated by the exact linear transport $B\, D21\, B^T$ plus a cheap, parity-respecting closure for the off-slice remainder R. The closure uses only the even algebra generated by C (the three monomials already written in the previous round) together with one or two additional even tensors that can be maintained at $O(n^2)$ cost (e.g. the diagonal of C^3 or a single learned scalar field of size n). Because the accumulator is only one matrix, its transport costs ≈ 1.4 units per layer; over the 11–12 layers that contain old content the extra bill is ≈ 15–20 units, bringing the total into the observed 133–165 window. The accuracy gain relative to pure young-source truncation (raw MSE ≈ 1e-6) comes from the fact that every layer still receives a nonzero, correctly transported odd contribution, satisfying the sensitivity map M2.

Confirming observable (local experiment, < 30 min): replace V29's old tier by a single D21 matrix transported exactly and closed with the three-monomial formula for R; measure final MSE. If the result lands inside $1.6–2.5\times10^{-8}$ the hypothesis is confirmed. If the error is still $> 5\times10^{-8}$, add the next even tensor suggested by a residual analysis of R and repeat. The same experiment also falsifies any claim that a more exotic object (characteristic function, full TAP expansion, etc.) is required.

Strongest argument that nothing cheaper remains: every scheme that retains a faithful representation of the deep odd third cumulant must either (i) transport objects whose size is $\Omega(n^2)$ (the slices themselves or equivalent response tensors) or (ii) keep a factorisation whose rank is forced by M3 to be several hundred. Both routes cost at least a few units per layer. The leaders' margin over the pure young tier is only 0–25 units for the whole network; that margin is precisely the cost of transporting one or two dense n×n matrices. Any proposal that claims zero marginal cost for the deep content contradicts either parity (T1), the measured spectra (M1), or the sensitivity map (M2).

Summary: the aggregate-response state of Q1 is elegant but almost certainly too expensive ($k \gtrsim 200$). The cheapest credible route consistent with all closed measurements is a single dense D21 accumulator plus an even closure for the off-slice remainder (Q3). That is also the most likely explanation of the leaders.